The Galois module structure of Deligne–Lusztig varieties for $\mathrm{G} = \mathrm{GL}_3$

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Introduction

In 1976, P. Deligne and G. Lusztig defined a pair of varieties, $\mathbf{Y}(w),\ \mathbf{X}(w)$ associated to a connected reductive group $\mathbf{G}$ defined over a finite field $\mathbb{F}_q$, with an endomorphism F such that its some positive power is a a power of the Frobenius endomorphism, and a Weyl group element $w \in W$. Furthermore, these varieties fit in a finite étale cover $\pi_w : \mathbf{Y}(w) \to \mathbf{X}(w)$, such that

$$\mathbf{X}(w) \simeq \mathbf{Y}(w)/\mathbf{T}^{wF},$$

where the covering group is given by the $\mathbb{F}_p-$ points $\mathbf{T}^{wF}$ of a torus.

We consider the ramified Galois covers $\overline{Y}(w) \to \overline{X}(w)$ of the smooth compactification of $X(w)$ obtained by the normalization of $\overline{X}(w)$ in $Y(w)$. Our goal is to calculate the Lefschetz numbers for the action of the covering group on the coherent cohomology of the structure sheaf of $\overline{Y}(w)$. These numbers determine the representation of $\mathbf{T}^{wF}$ on the alternating sum of these cohomology groups. We will describe results giving explicitly these Lefschetz numbers in the cases of Deligne-Lusztig surfaces for $\mathbf{G}^F = \mathrm{GL}_3(\mathbb{F}_p)$ and $\mathbf{G}^F = \mathrm{U}_3(\mathbb{F}_p)$.

Deligne–Lusztig varieties

Let $\mathbf{G} = \mathrm{GL}_n$ be defined over $k = \mathbb{F}_q$, where q is some power of the prime p. Let F be an endomorphism such that F^δ is a Frobenius endomorphism for some $\delta \geq 1$. Let $\mathbf{T} \subseteq \mathbf{G}$ be the subgroup of diagonal matrices, and let $\mathbf{B} \subseteq \mathbf{G}$ be the subgroup of upper triangular matrices, Let $W = N_{\mathbf{G}}(\mathbf{T})/\mathbf{T}$ be the Weyl group which is isomorphic to S_n. Note that we have the Bruhat decomposition into open *Bruhat cells* $\mathbf{G} = \bigsqcup_{w \in W} \mathbf{B}\dot{w}\mathbf{B}$.

The homogeneous space $\mathscr{X} := \mathbf{G}/\mathbf{B}$ can be identified with the full flag variety

$$\mathrm{Fl} = \{(0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_{n-1} \subset V_n = k^n) | \dim_k(V_i) = i\}$$

We cut out a locally closed subvariety $\mathbf{X}(w)$ of Fl called the **Deligne–Lusztig variety** defined by

$$\mathbf{X}(w) \simeq \{\, g\mathbf{B} \in \mathbf{G}/\mathbf{B} \mid g^{-1}F(g) \in \mathbf{B}\dot{w}\mathbf{B} \,\}$$

Let $\mathbf{U}$ be the subgroup of unipotent upper triangular matrices, the unipotent radical of $\mathbf{B}$. Then

$$\mathscr{Y} := \mathbf{G}/\mathbf{U} \to \mathscr{X} = \mathbf{G}/\mathbf{B}$$

is a $\mathbf{T}$-torsor. Note that there is a natural action of the finite group $\mathbf{G}^F$ on $\mathscr{Y}$.
We define similarly a locally closed subvariety $\mathbf{Y}(w) \subset \mathbf{G}/\mathbf{U}$ of $\mathscr{Y}$ also called the **Deligne-Lusztig variety (scheme*)**

$$\mathbf{Y}(w) = \{\, g\mathbf{U} \in \mathbf{G}/\mathbf{U} \mid g^{-1}F(g) \in \mathbf{U}\dot{w}\mathbf{U} \,\} \subseteq \mathscr{Y}.$$

Then, $\mathbf{Y}(w) \to \mathbf{X}(w)$ is a $\mathbf{T}^{wF}$-torsor, where

$$\mathbf{T}^{wF} = \{\, t \in \mathbf{T} \mid \dot{w}F(t)\dot{w}^{-1} = t \,\}.$$

Let $\overline{X}(w)$ denote the smooth compactification of $X(w)$. Bonnafé and Rouquier constructed the normalization $\overline{\mathbf{Y}}(w)$ of $\overline{\mathbf{X}}(w)$ in $\mathbf{Y}(w)$ and proved the following theorem that the normal variety $\overline{\mathbf{Y}}(w)$ is equipped with an action of $\mathbf{T}^{\mathbf{w}F}$ extending the action on $\mathbf{Y}(w)$ and such that $\overline{\pi}_{\mathbf{w}}$ induces an isomorphism $\overline{\mathbf{Y}}(w)/\mathbf{T}^{\mathbf{w}F} \xrightarrow{\sim} \overline{\mathbf{X}}(w)$.

$$\begin{array}{ccc} \mathbf{Y}(w) & \hookrightarrow & \overline{\mathbf{Y}}(w) \\ \downarrow{\scriptstyle \pi_w} & & \downarrow{\scriptstyle \overline{\pi}_w} \\ \mathbf{X}(w) & \hookrightarrow & \overline{\mathbf{X}}(w) \end{array}$$

Some examples

- Let $\mathbf{G} = \mathrm{GL}_n \,/\, \overline{\mathbb{F}}_p,\ F = \mathrm{Frob}_p$. Let $w = (1\ 2\ \ldots\ n) \in S_n$ to be the Coxeter element. Then,

$$X(w) = \mathbb{P}^{n-1} \setminus \{\mathbb{F}_p - \text{rational hyperplanes}\} \rightsquigarrow \textit{The Drinfeld half-space}$$

$$Y(w) = \left\{ (x_1, \ldots, x_n) \in \mathbb{A}^n_{\overline{\mathbb{F}}_p} : (-1)^{n-1} \det \left(x_i^{p^{j-1}} \right)^{p-1}_{1 \leq i,j \leq n} - 1 = 0 \right\}$$

  Specifically, for $n = 2$, we get one of the irreducible components of $Y(w)$ to be the famous **Drinfeld curve** $xy^p - x^p y = 1$.
- Let $\mathbf{G} = \mathrm{GL}_3 \,/\, \overline{\mathbb{F}}_p,\ F = J \cdot \mathrm{Frob}_p^{-t} \cdot J^{-1}$, where $J = \mathrm{antidiag}(1, \ldots, 1)$. Let $w = (1\ 2\ 3) \in S_3$ to be the Coxeter element. Then,

$$X(w) = \{([x];[y]) \in \mathrm{Fl} \subset \mathbb{P}^2_{[x]} \times \mathbb{P}^2_{[y]} \mid y_0 y_2^p - y_1^{p+1} + y_2 y_0^p \neq 0\}$$

Galois module structures

One of the central problems in the subject of "geometric" Galois module theory is as follows:

When G is a finite group acting on a projective scheme Y over $\mathrm{Spec}(k)$ and $\mathcal{F}$ is a $G-$equivariant coherent sheaf of $\mathcal{O}_Y-$modules, the sheaf cohomology groups $H^i(Y, \mathcal{F})$ are G-modules, and one asks if its equivariant Euler characteristic

$$\chi(Y, \mathcal{F}) := \sum_i (-1)^i H^i(Y, \mathcal{F})$$

can be calculated using a bounded complex of finitely generated free $k[G]-$modules.

We look at the virtual Galois module inside the Grothendieck group $K_0(k[G])$.

$$\mathcal{E} := [\chi(Y, \mathcal{F})] = \sum_i (-1)^i [H^i(Y, \mathcal{F})]$$

If $k[G]$ is semi-simple, (this is true, for instance, if we have *"tame ramification", i.e., if* $\mathrm{char}(k) \nmid |G|$), then any element in this Grothendieck group is uniquely and entirely determined by its Brauer character, $\chi_{Br} : G \to \mathbb{C}$ in the following way:

The Lefschetz number $L(\tau, \mathcal{F})$ of an element $\tau \in G$ is the Brauer character χ_{Br} evaluated at τ. i.e,

$$\chi_{Br}(\tau) = L(\tau, \mathcal{F}) = \sum_i (-1)^i \mathrm{Tr}(\tau | H^i(Y, \mathcal{F}))$$

Theorem 1 (The split case)

Let $\mathbf{G} = \mathrm{GL}_3 \,/\, \mathbb{F}_p,\ F = \mathrm{Frob}_p,\ w = (1\ 2\ 3)$. Also let ζ be a fixed primitive $(N-$th complex root of unity where $N := p^2 + p + 1$. Suppose $\mathbf{T}^{wF} \simeq \mathbb{Z}/(p^3 - 1)\mathbb{Z} = \langle \tau \rangle$. Then,

the Lefschetz number for an irreducible component of the corresponding Deligne–Lusztig variety $\overline{Y}(w)$ is given by

$$L(\tau^{p-1}, \overline{Y}_\ell) = N(p+1)\left(\frac{\zeta(1-\zeta^{p-1})}{(1-\zeta)^2(1-\zeta^p)} + \sum_{k=2}^{p} \frac{1}{1-\zeta^k} \right) + \frac{N\zeta}{(\zeta-1)^2} + \frac{pN\zeta^p}{(\zeta^p-1)^2}$$

where $\overline{Y} = \bigsqcup_{\ell \in \mathbb{F}_p^\times} \overline{Y}_\ell$ is a decomposition into finitely many irreducible subvarieties.

Theorem 2 (The non–split case)

Let $\mathbf{G} = \mathrm{GL}_3 \,/\, \mathbb{F}_p,\ F = J \cdot (\mathrm{Frob}_p)^{-t} \cdot J^{-1}$, where $J = \mathrm{antidiag}(1, -1, 1),\ w = (1\ 2\ 3)$. Also let ζ be a fixed primitive $(p^2-1)-$th complex root of unity. Then, $\mathbf{T}^{wF} \simeq \mathbb{Z}/(p+1)\mathbb{Z} \times \mathbb{Z}/(p^2-1)\mathbb{Z} = \langle \tau^{p-1} \rangle \times \langle \tau \rangle$ and,

the Lefschetz number for an irreducible component of the corresponding Deligne–Lusztig variety $\overline{Y}(w)$ is given by

$$L(\tau, \overline{Y}_\ell) = \frac{\zeta(p+1)(p-2)(2p+1-\zeta)}{2(1-\zeta)^2} - \frac{(p^3+1)\zeta}{(1-\zeta)(1-\zeta^p)} - \frac{(p+1)\left(p-2+3p\zeta^p\right)}{2(1-\zeta^p)^2}$$

where $\overline{Y} = \bigsqcup_{\ell \in \{1, \ldots, p+1\}} \overline{Y}_\ell$ is a decomposition into finitely many irreducible subvarieties.

Proof Strategy

- We work (étale) locally to resolve the singularities of $\overline{Y}$ and construct a smooth model $\widetilde{Y}'$ with an action of the Galois group T^{wF} extending that of it on $\overline{Y}$.
- We find that the local calculations leads us to the realm of toric geometry, and the **Hirzebruch-Jung algorithm** for finding the minimal (toric) resolution of toric surfaces gives us all the tools required to apply the *Lefschetz-Riemann-Roch theorem* to compute the Lefschetz numbers.

Equivariant resolutions and the smooth toric model

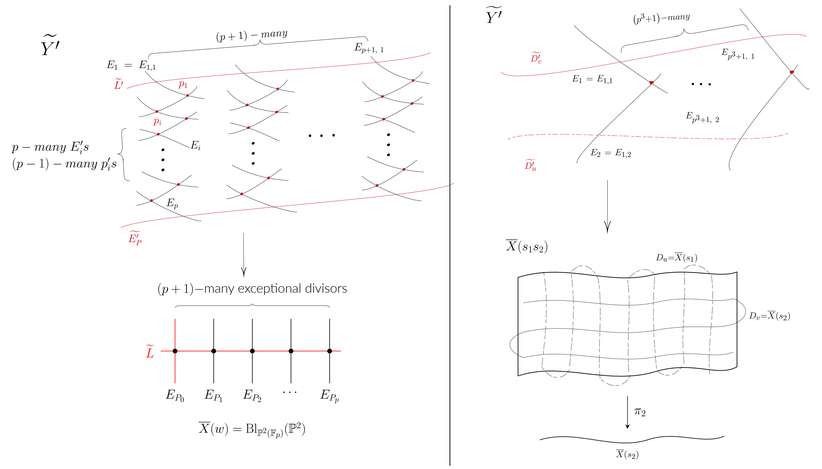


Figure 1. Resolutions of the split ($G^F = \mathrm{GL}_3(\mathbb{F}_p)$) case on the left and the non-split ($G^F = \mathrm{U}_3(\mathbb{F}_p)$) case on the right.

References

[1] Cédric Bonnafé and Raphaël Rouquier. Compactification des variétés de Deligne-Lusztig. *Annales de l'Institut Fourier*, 59(2):621–640, 2009.

[2] D.A. Cox, J.B. Little, and H.K. Schenck. *Toric Varieties*. Graduate studies in mathematics. American Mathematical Society, 2011.

[3] P. Deligne and G. Lusztig. Representations of Reductive Groups Over Finite Fields. *The Annals of Mathematics*, 103(1):103, January 1976.

[4] Ulrich Görtz and Stefan Schröer. Deligne-Lusztig varieties whose canonical divisors have negativity, May 2026. arXiv:2605.02522 [math.AG] version: 1.